

PSimPy - Predictive and probabilistic simulation with Python



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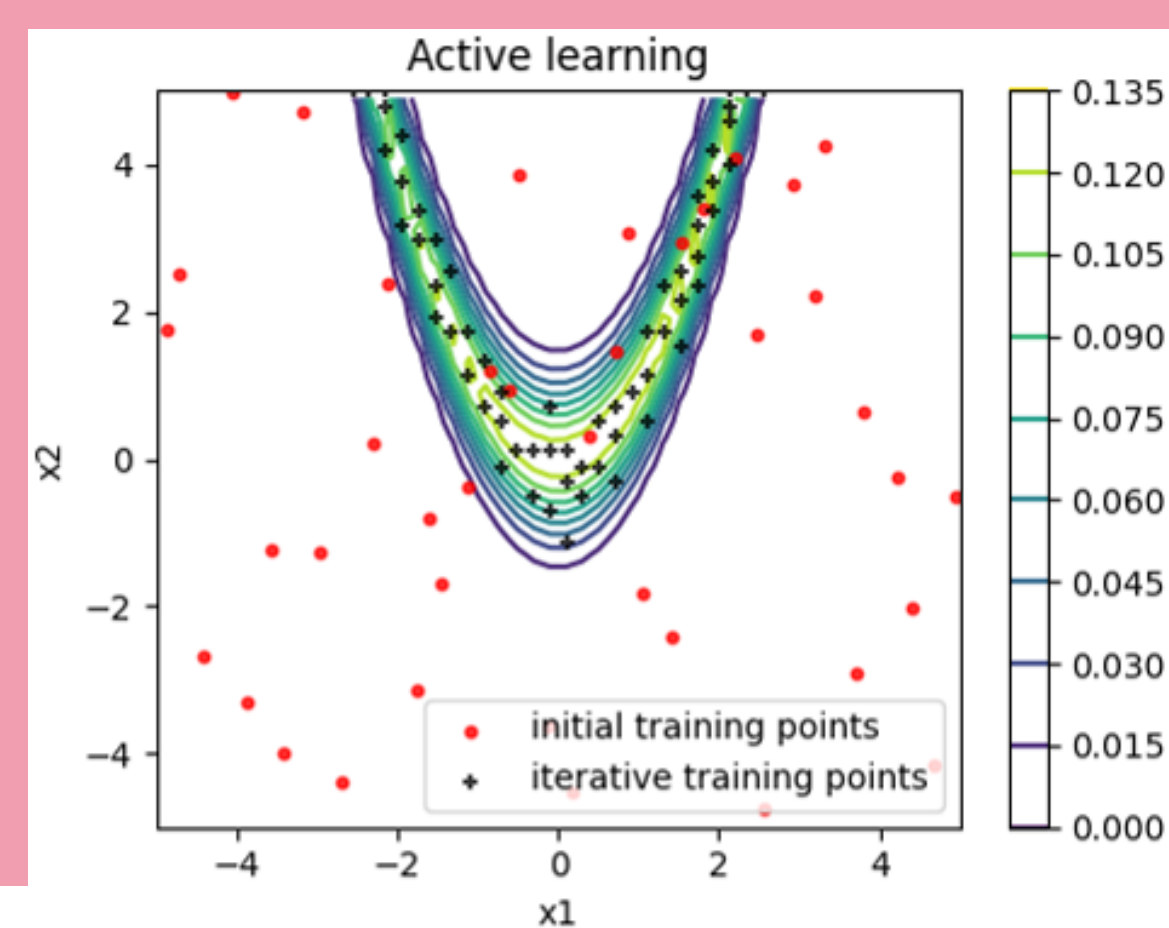
Gaussian Process, Multiphysics Simulation

Summary

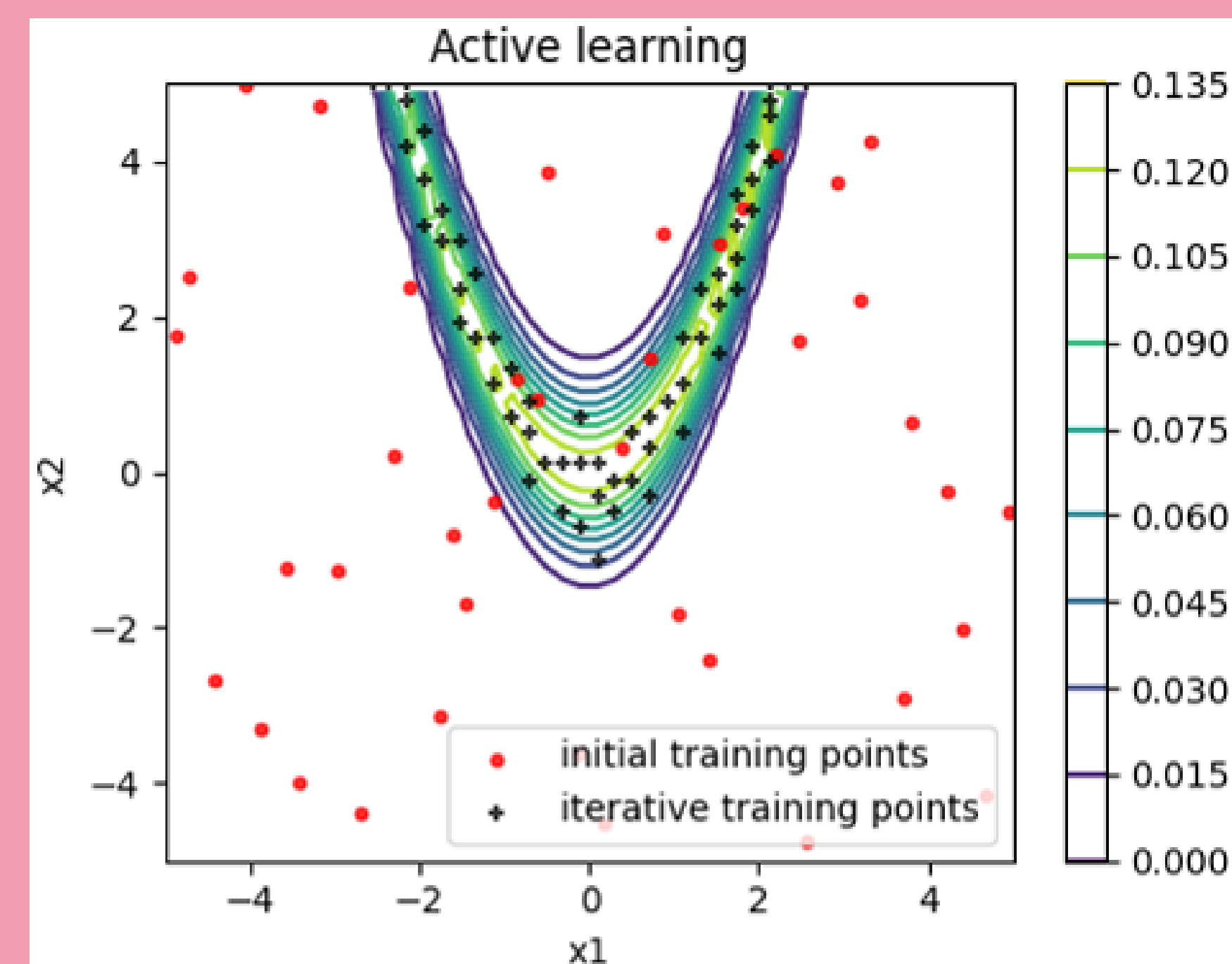
Train an GP Emulator and save compute time!

	training	evaluation	2500 evals
simulator	-	32 min	56 days
emulator	4.5 days	0.05 sec	2 min

- Python wrapper
- Robust GP Emulation
- Lightweight Substitute



Software in Action



Identify key drivers. Calibrate with data.

- Run sensitivity analysis using Sobol indices
- Sample efficiently using Saltelli, LHS, or Metropolis Hastings

Want to build unified,
robust and sustainable
UQ workflows?

PSimPy^[1]



Bibliography

- [1] H. Zhao, A. Yildiz, N. Bagherinejad, and J. Kowalski, “PSimPy: GP emulation-based sensitivity analysis, uncertainty quantification and calibration of landslide simulators,” in *14th International Conference on Applications of Statistics and Probability in Civil Engineering, ICASP14*, Dublin, Ireland, 2023.